

DR. ZAHRA MOKHTARI

Data Scientist | Statistical Physicist

✉ emokhtari89@gmail.com  zahramokhtari  @zmokhtari89  Berlin, Germany



ABOUT ME

A data scientist with a PhD in statistical physics (**my peer-reviewed publications**), I build end-to-end solutions from data to deployment in production environments. My passion for finding patterns in data is backed by strong cross-disciplinary communication skills.

EXPERIENCE

Data Scientist

AUTO1 Group – AutoPricing Team

📅 2025 Nov. – 2026 Mar

📍 Berlin, Germany

- Implemented and deployed weighted aggregation of four models based on feature-space distance to predict sell price, improving sell rate and gross profit (A/B tested).
- Built LightGBM model with custom Numba-accelerated loss function penalizing cheap-car errors more aggressively, boosting performance in key price categories.
- Maintained end-to-end ML pipeline; monitored model performance post-retraining, managed releases to production, and resolved AWS Step Functions/Jenkins build failures.

Parental Leave

📅 2019–2020, 2022–2023

📍 Berlin, Germany

Postdoctoral Researcher

Freie Universität Berlin (FU) – Department of Mathematics and Computer Science

📅 2020 Jul. – 2024 Nov.

📍 Berlin, Germany

- Implemented agent-based simulations (C++/CUDA) to model pattern formation in colonies of ants, with up to ~400K agents
- Developed and applied image processing pipelines to quantify large-scale spatial patterns from simulation data.
- Mentored 50+ students in Physics, Math, Scientific Computing.
- Lectured on statistical mechanics and computational methods.

PhD/Postdoctoral Researcher

University of Göttingen – Institute for Theoretical Physics

📅 2013 Oct.. – 2019 Mar.

📍 Göttingen, Germany

- Built parallelized simulations of stochastic models of bacterial migration, processing TB-scale trajectory data on HPC clusters.
- Managed cross-disciplinary collaboration (math/biology), to derive insights from simulations.
- Published first-author papers in peer-reviewed journals and presented results at international conferences.

SKILLS

Languages  Python  SQL  C/C++  ML & DL

 scikit-learn  TensorFlow  Data Engineering

& Tools  Git  Docker  MLflow  Streamlit

 AWS  GCP  FastAPI  Data Analysis

 Statistical Analysis  Matplotlib  Tableau

PROJECTS

Unsupervised Music Pattern Discovery

📅 2025

📍 Le Wagon

- Applied unsupervised ML (K-means) to audio data for pattern extraction.
- Deployed as a Dockerized pipeline on Google Cloud; Streamlit visualization.

Agent-Based Modeling of Biological Patterns

📅 2022

📍 Freie Universität Berlin

- Built a GPU-accelerated simulation, with analysis in Python (NumPy/scikit-image).

EDUCATION

Certificate in Data Science and AI

Le Wagon

📅 Jan. 2025 – Mar. 2025

📍 Berlin, Germany

- Hands-on ML, DL (DNNs, RNNs, LLMs), & engineering (MLflow, FastAPI).
- Pitched and led "Music Pattern Discovery" project (team of 4).

PhD in Statistical Physics

University of Göttingen

📅 2013 – 2018

📍 Göttingen, Germany

- Thesis – Bacteria in Heterogeneous Media

M.Sc. in Physics

Simon Fraser University

📅 2011 – 2013

📍 Vancouver, Canada

- Thesis – Quantum Ising chains in fields